Average Time Spent by Users on Social Media

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Introduction

This representation examines a dataset that tracks user engagement on social media based on age, gender, income, profession, demographics, interests, location, platform, indebt, home,car owner.
The primary objective is to explore these variables, build predictive models, .and derive meaningful insights about social media consumption

Data Exploration

The dataset contains multiple attributes related to users' social media usage, including:

age: User's age (ranging from 18 to 64).

gender: User's gender identity (Male, Female, Non-binary)

demographics: The area type where the user resides (Urban, Suburban, Rural).

interests: The user's primary interests or hobbies.

location: The country of residence.

platform: The social media platform most used.

Data Exploration

profession: The user’s occupation .

income: The annual income of the user

indebt: Whether the user is in debt (True/False).

homeowner: Whether the user owns a home (True/False) m

owns_cars: Whether the user owns cars (True/False).

Exploratory Data Analysis (EDA)

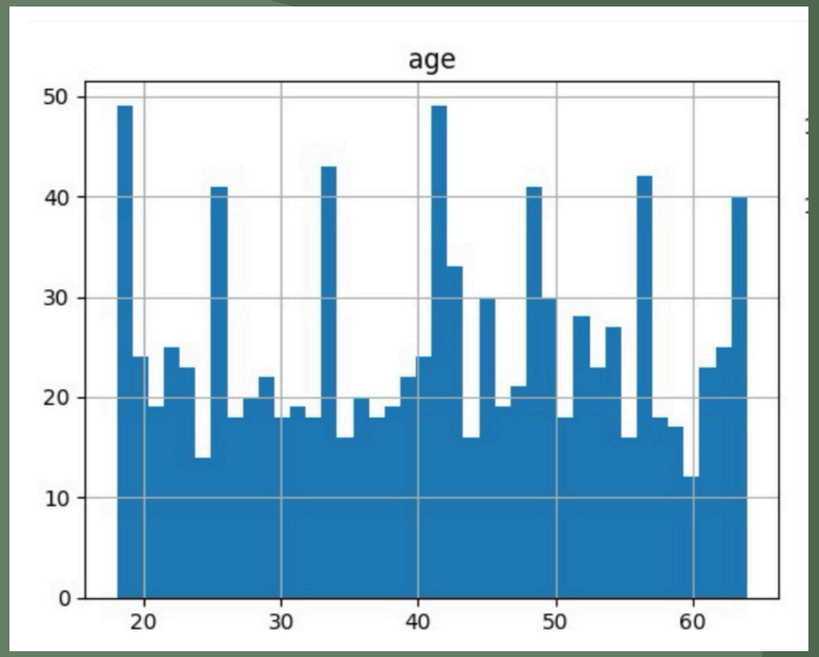
Most users belong to middle-aged groups
Income variations influence social media usage duration
Users are distributed across different living environments

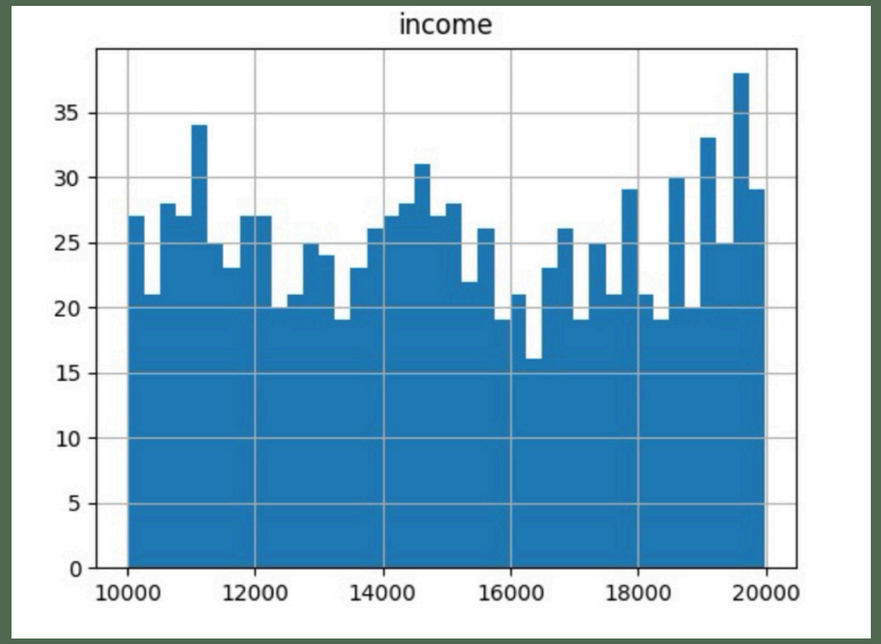
Modeling Process:

Regression Model: Examines the relationship between income and time spent on social media)
Feature Selection: Key factors include age, gender, platform, profession, and financialstatus
Model Tuning: Optimized parameters to improve accuracy

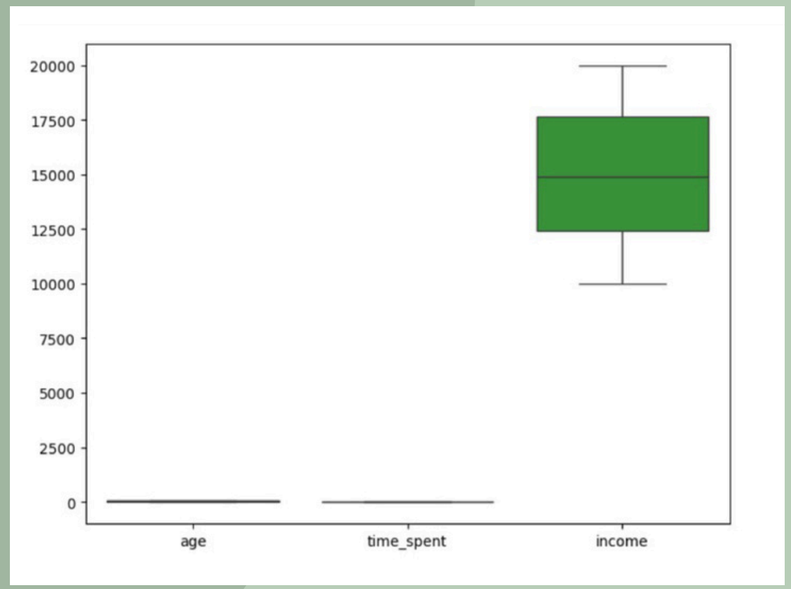
Evaluation & Visual Representations

- **Age**: shows variability in age distribution
- **Time spent** : Indicates an equal distribution across categories
- **Income** :Displays variability in income levels

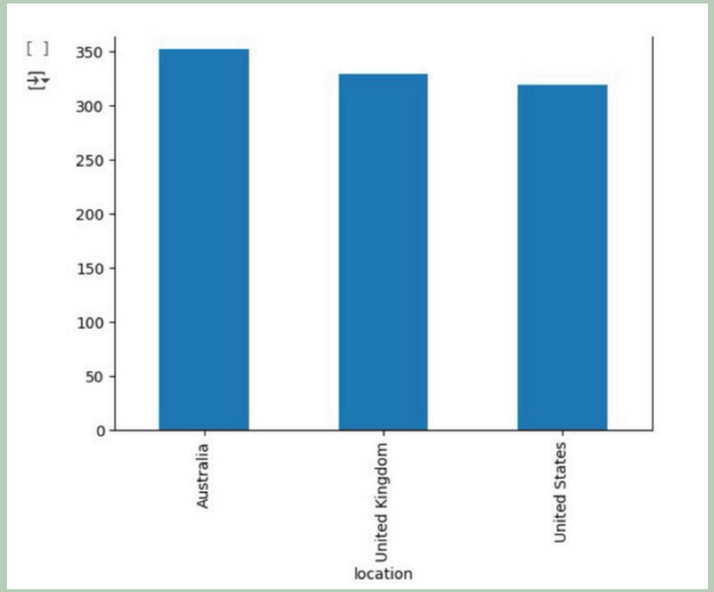




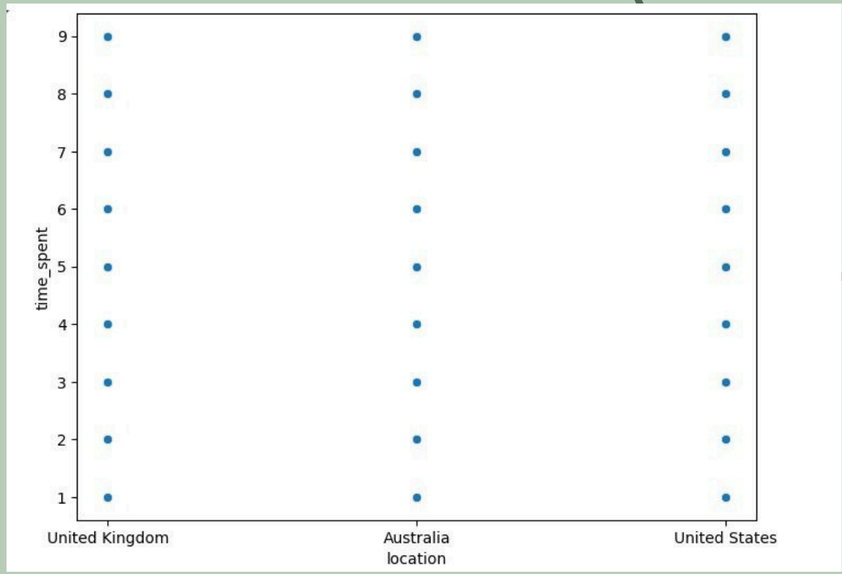
Box plot :
Illustrates key statistics (min, quartiles, max) for age, time spent, and income, highlighting data spread and outliers.



Bar Chart by Location :
Compares the number of ,individuals from Australia the UK, and the US, showing a similar count across these countries

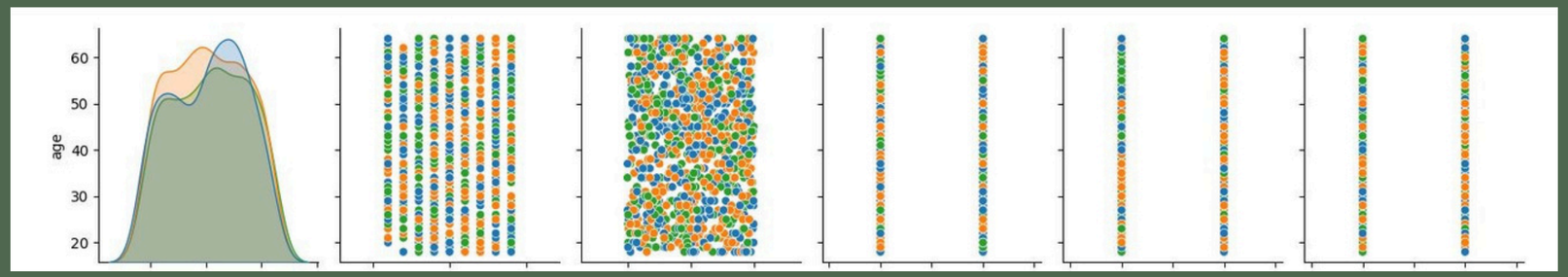


Scatter Plot of Time Spent by Location* :
-
Displays points representing individuals from the UK, Australia, and the US based on .their time spent. Each dot represents a data point



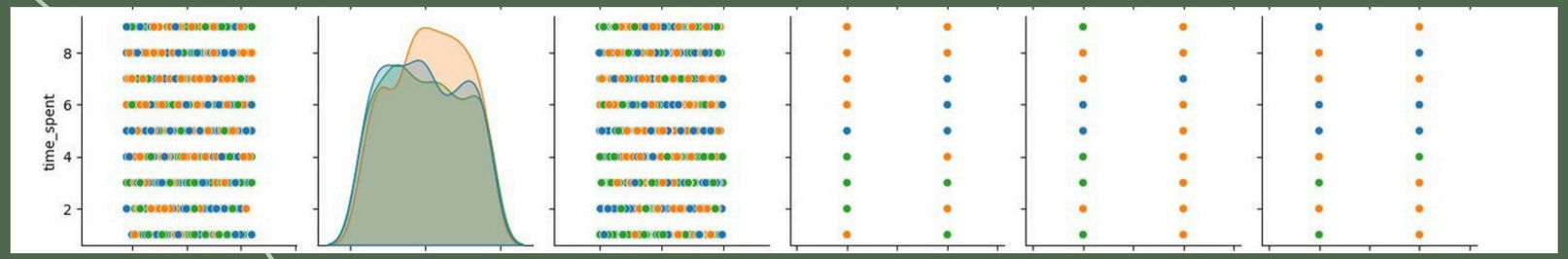
Age Distribution and Related Graphs*

Shows the distribution of ages with a density plot. The following graphs likely compare age data -
.across different categories or locations



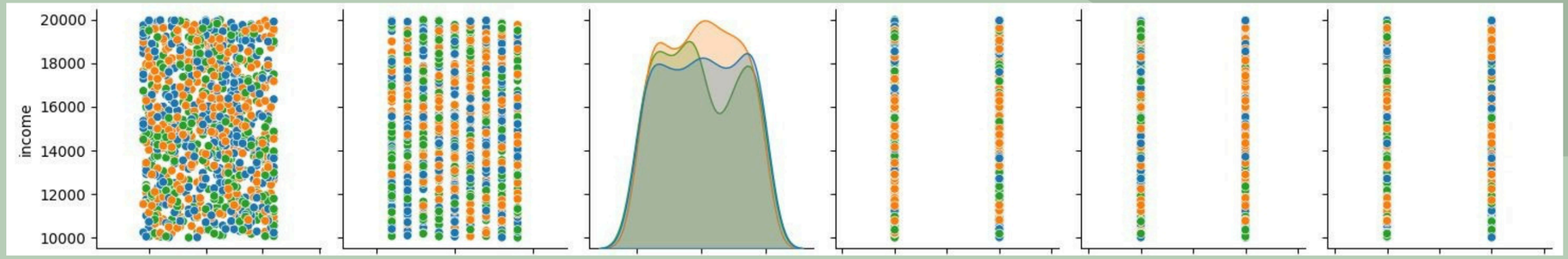
Time Spent Distribution and Related Graphs*

Displays the distribution of time spent with a density plot. Similar to the previous image, the -
.following graphs likely compare time spent across categories



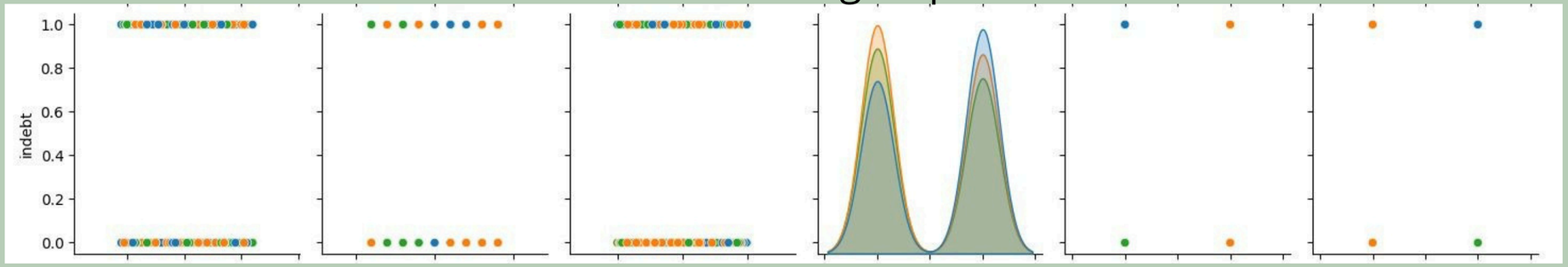
Income Distribution and Related Graphs*

Illustrates income distribution with a density plot. The subsequent graphs may also compare income across - .various categories



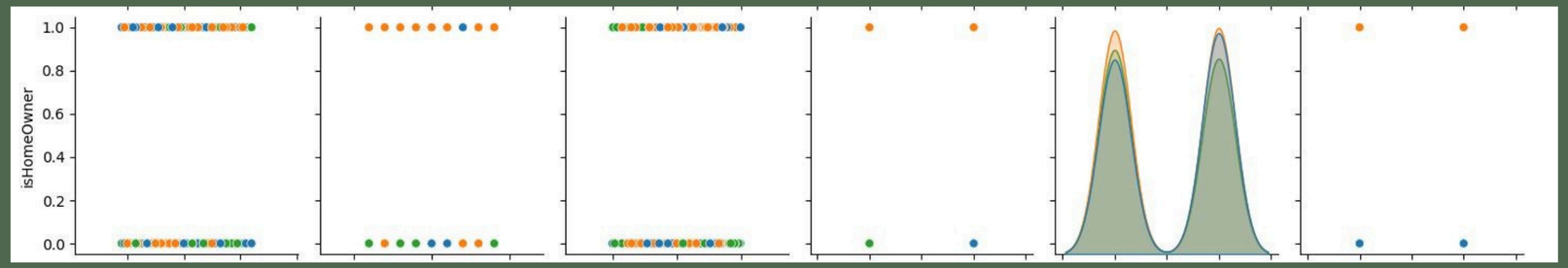
Indebtedness Distribution and Related Graphs*

Shows the distribution of indebtedness with a density plot, indicating how indebtedness varies across - different groups



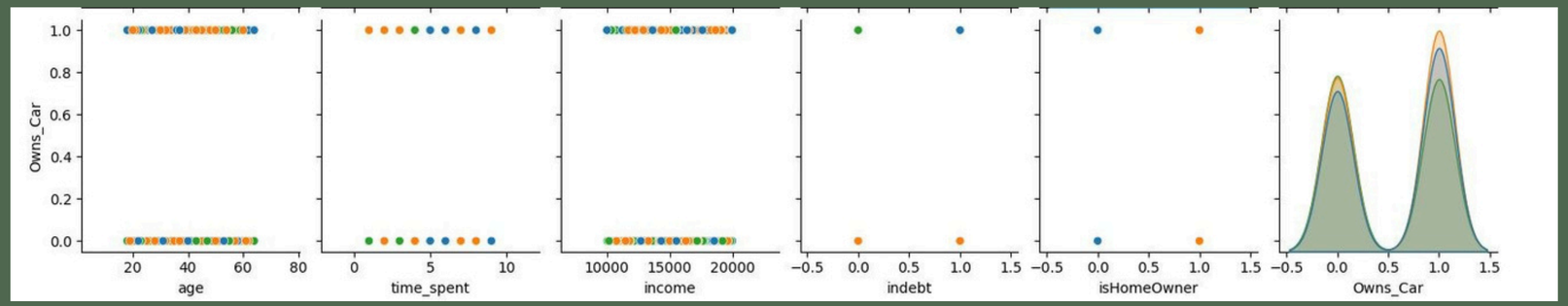
Homeownership Distribution and Related Graphs*

Displays homeownership status with a density plot, demonstrating the proportion of homeowners - .versus non-homeowners



Car Ownership Distribution and Related Graphs*

Illustrates car ownership status with a density plot, indicating the relationship between various factors - .like age, income, etc



Legend for Locations*
Provides a color-coded legend indicating - which colors represent individuals from the UK, .Australia, and the US

Conclusions

- Age, income, and platform choice significantly influence engagement.

- Middle-aged and urban users are the most active, while higher-income users tend to engage less.
- Interests and lifestyle choices also impact time spent on social media.
- Businesses can use these insights to create personalized marketing strategies and enhance user engagement.

Thank you!